Prelab: Introduction to Pointers

Overview

This prelab is designed to prepare you for the upcoming lab work on pointers. It covers the fundamental concepts of memory addresses, pointer declaration, dereferencing, and basic pointer arithmetic.

Part 1: Conceptual Questions

1. **What does the `&` operator do in C?**

- a) Returns the value of a variable
- b) Returns the memory address of a variable
- c) Declares a pointer
- d) Dereferences a pointer

2. **What does the `*` operator do when placed before a pointer variable (e.g., `*ptr`)?**

- a) Returns the address of the pointer
- b) Declares a new variable
- c) Accesses the value stored at the address held by the pointer
- d) Multiplies the pointer by itself

3. **If `int x = 10;` and `int *p = &x;`, what is the type of `p`?**

- a) `int`
- b) `int *`
- c) `&int`
- d) `address`

4. **In a 64-bit system, what is the typical size of a pointer variable?**

- a) 2 bytes
- b) 4 bytes
- c) 8 bytes
- d) Depends on the type it points to

5. **Explain the difference between `ptr++` and `(*ptr)++`.**

- (Write your answer here)

6. **What is the value of a pointer initialized to `NULL`?**

- a) 0
- b) 1
- c) Random garbage value
- d) The address of the NULL variable

7. **Can a `void` pointer be dereferenced directly?**

- a) Yes
- b) No, it must be cast first
- c) Only if it points to an integer
- d) Only in C++

Part 2: Code Tracing

Determine the output of the following code snippets.

Snippet 1

```
int a = 5;
int *p = &a;
*p = 10;
printf("%d", a);
```

Output: ________

Snippet 2

```
int arr[] = {10, 20, 30};
int *ptr = arr;
ptr++;
printf("%d", *ptr);
```

Output: ________

Snippet 3

```
int x = 50;
int *p1 = &x;
int **p2 = &p1;
**p2 = 100;
printf("%d", x);
```

Output: ________

Snippet 4

```
int arr[] = {10, 20, 30, 40};
int *p1 = arr;
int *p2 = arr + 2;
printf("%ld", p2 - p1);
```

Output: ________

Snippet 5

```c
int arr[] = {50, 60};
int *p = arr;
int res = *p++;
printf("res=%d, *p=%d", res, *p);
```

Output: ________

Part 3: Spot the Error

Identify the error in each of the following snippets.

Snippet 1

```c
int *ptr;
*ptr = 10;
```

Error: ____

Snippet 2

```c
float pi = 3.14;
int *p = &pi;
```

Error: ____

Snippet 3

```c
int a = 5;
int *p = a;
```

Error: ____

Part 4: Simple Coding

Write short code snippets to solve the following.

1. **Declare and Initialize:** Declare an integer `count` initialized to 0. Declare a pointer `p_count` that points to `count`.
2. **Update via Pointer:** Using the pointer `p_count` from above, increment the value of `count` by 5.

3. **Print Address:** Print the memory address of `count` using the `%p` format specifier.
4. **Double Pointer:** Declare a double pointer `pp_count` that points to `p_count`. Use `pp_count` to print the value of `count`.